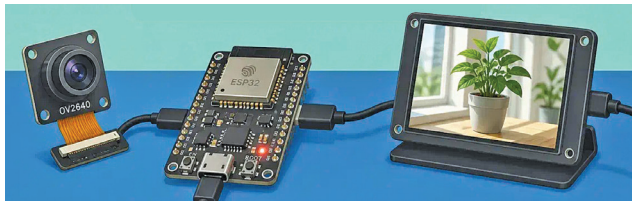
India takes edge computing into orbit

Indian space startup TakeMe2Space has launched MOI-1A, a sub-50kg satellite designed to run AI workloads in orbit instead of sending raw data to Earth. The spacecraft, launched on SpaceX's Transporter-18 on October 1, uses Nvidia Orin NX processors for onboard edge computing within a reported 150W power budget. Its containerised architecture allows users to upload AI models for satellite-image and sensor-data inference. By processing data in space and downlinking only results, the system can reduce communication bandwidth requirements and deliver insights faster. The approach could enable data-intensive Earth observation and other orbital edge applications.

Arduino gets advanced camera support

Espressif has added MIPI-CSI camera support to the Arduino ecosystem through its new ESP_Video library, enabling camera capture on ESP32-P4 boards without a separate



The library wraps Espressif's ESP-IDF video component in C++ classes, exposing MIPI-CSI and DVP camera interfaces as separate video device nodes (Image: Espressif)

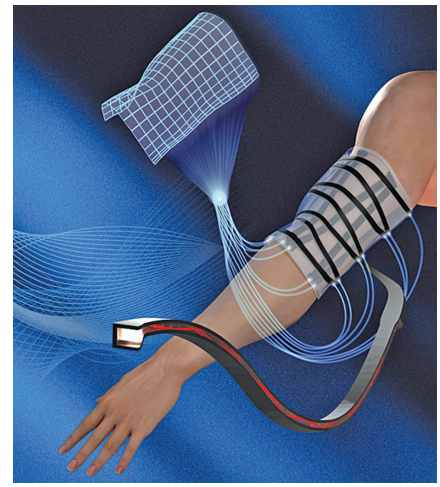
ESP-IDF project. The library wraps Espressif's video framework in C++ classes and supports MIPI-CSI and DVP interfaces; the ESP32-P4 can run both paths simultaneously, while the ESP32-S3 supports DVP. Its V4L2-style API simplifies camera prototyping for embedded vision, monitoring, and image-processing applications. It requires Arduino-ESP32 Core 3.3.10 or later and can use PSRAM for frame buffering, lowering software barriers for makers and embedded developers.

Wireless power for stretchable implants

Researchers at the Institute for Basic Science and Seoul National University engineered a wearable transmitter with a self-oscillating, parity-time-symmetric circuit and a stretchable liquid-metal receiver. It can keep soft medical implants powered despite movement, stretching, and misalignment, addressing a key limitation of conventional inductive charging. The system maintained more than 50 per cent power-transfer efficiency under 30 per cent strain and 30mm misalignment. Tests in rabbit and pig models demonstrated wireless cardiac pacing and tachyarrhythmia termination. The approach could enable battery-free or battery-reduced implants, although human testing and long-term safety studies remain necessary.

A flexible sheet that tracks deformation

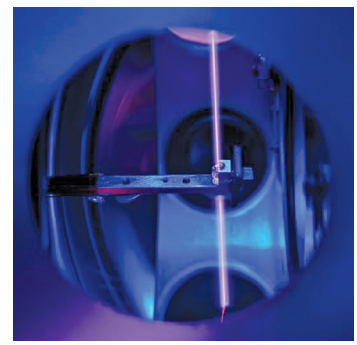
Engineers at the Massachusetts Institute of Technology (MIT), USA, have developed a flexible sheet that tracks bending and twisting, reconstructing its three-dimensional shape in near real time. Embedded zig-zag optical fibres detect deformation through changes in transmitted light, which sensors measure and an algorithm converts into a digital model. Tests showed reconstruction errors below 4mm, even when some fibres were damaged or disconnected. The technology could help physical therapists monitor rehabilitation progress and enable gesture-based control of robots and video game characters, advancing flexible sensing systems.



In physical therapy, you often need to gather patient data on how they are moving and stretching their arm during rehabilitation. Surface shape sensing could be very useful (Image: Courtesy of the researchers)

A self-correcting nuclear clock

Physicists at TU Wien in Vienna, Austria, have demonstrated a nuclear clock that automatically corrects laser-frequency drift using thorium-229 nuclei, enabling continuous operation for more than 24 hours. The system embeds the nuclei in a calcium fluoride crystal and monitors light absorption to detect frequency deviations, feeding corrections back to the laser. Researchers reported frequency instability approaching one part in a quadrillion over a day. The technology could advance precision measurements, tests of fundamental physics, and future timekeeping, although further improvements are needed before practical deployment.



The thorium crystal and the laser beam (Image: TU Wien)